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Case Report

Variable Effects of Glucagon Like Peptide-1 Receptor Agonists on Body Composition in Older Women

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ABSTRACT

Background/Objective: Glucagon-like peptide-1 receptor agonists (GLP-1 RAs) improve glycemic control, promote weight loss, and enhance cardiovascular outcomes. However, their effects on lean mass in older adults are unclear. This case report describes how GLP-1 RAs influenced body composition in 2 older women.

Case Report: We measured the body compositions of 2 older women with obesity, who were unable to lose weight through lifestyle changes alone, before and after taking GLP-1 RAs. The first patient (65 years old) received weekly subcutaneous injections of tirzepatide. She had a moderate reduction in body mass index over 4 months, with a modest reduction in fat mass and a substantial reduction in lean mass. The second patient (79 years old) received weekly subcutaneous injections of semaglutide and had a moderate reduction in body mass index over 6 months, with a substantial reduction in fat mass and a modest gain in lean mass. The patient with greater muscle loss was less physically active and tracked food intake less precisely than the patient with less muscle loss.

Discussion: Differences in lean mass loss between patients may be influenced by exercise and calorie counting. These findings highlight the importance of lifestyle modifications for preserving lean mass. However, because only 2 patients were studied, the results are entirely anecdotal and therefore cannot be generalized to larger populations.

Conclusion: Clinicians should consider recommending individualized exercise programs and dietary strategies for older patients taking GLP-1 RAs. Routine assessment of body composition may help guide interventions and optimize clinical outcomes.

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Introduction

Obesity, a chronic disease characterized by the excess accumulation of adipose tissue, can impair functional mobility and cause various comorbid conditions. Among Americans older than 60 years, 41.5% are classified as obese by the American Heart Association, which defines obesity as a body mass index (BMI, calculated as weight in kilograms divided by height in meters squared) of 30 or higher.[1] Obesity has been linked to hypertension, coronary artery disease, heart failure, hyperlipidemia, diabetes, obstructive sleep apnea, and many more conditions that shorten lifespan.[2] Traditional efforts to treat obesity include lifestyle modifications such as exercise and diet.[3] However, older patients with obesity often have mobility-restricting comorbid conditions—such as arthritis, cardiovascular disease, pulmonary dysfunction, and sarcopenia—that make exercise difficult.[4-7] Glucagon-like peptide-1 receptor agonists (GLP-1 RAs) can facilitate weight loss without the need for lifestyle modifications by increasing satiety and reducing food intake. GLP-1 RAs bind to pancreatic beta cells to increase insulin secretion, reduce glucagon secretion from pancreatic alpha cells, and delay gastric emptying, all of which contribute to postprandial glycemic control and feelings of satiety.[8] The SURMOUNT-1 (SURMOUNT-1 was a phase 3, double-blind, placebo-controlled trial evaluating tirzepatide [a dual GIP/GLP-1 receptor agonist] for weight management in adults without diabetes who had overweight or obesity. Over 72 weeks, participants received weekly tirzepatide (5 mg, 10 mg, or 15 mg) or placebo alongside lifestyle

Abbreviations: BMI, body mass index; GLP-1 RA, glucagon-like peptide-1 receptor agonist.

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intervention. The study showed significant, dose-dependent weight loss — up to about 21% of body weight at the highest dose — compared with ~3% with placebo, establishing tirzepatide as a powerful option for obesity treatment) trial[9] showed that tirzepatide can lead to 15% to 21% weight loss over 72 weeks in obese patients without diabetes, depending on the dosage regimen. In a similar population, the STEP 1 (STEP-1 was a phase 3, randomized, placebo-controlled trial testing semaglutide 2.4 mg once weekly in adults with overweight or obesity but without diabetes. Conducted over 68 weeks, it combined medication with lifestyle intervention and demonstrated an average 14.9% weight reduction versus 2.4% with placebo. The trial provided the pivotal evidence leading to FDA approval of semaglutide 2.4 mg [Wegovy] for chronic weight management) trial[10] reported mean weight loss of 15% to 21% over 68 weeks with semaglutide. However, it is important to determine whether this weight reduction is due to fat loss or muscle loss. Cross-sectional studies spanning 25 years have shown that most muscle groups decrease in mass by 0.28% to 0.58% per year.[11] Increased muscle loss due to GLP-1 RA therapy can exacerbate existing sarcopenia in older patients and thus increase the risk of falls, hospitalizations, functional decline, and poor quality of life.[12-14] Few studies have investigated how GLP-1 RAs affect the body composition of older patients. This study aimed to describe body composition changes in 2 women aged 65 years and older who were taking GLP-1 RAs.

Highlights

- Glucagon-like peptide-1 receptor agonists (GLP-1 RAs) may cause muscle loss in older adults
- Exercise and food tracking may reduce muscle loss in patients taking GLP-1 RAs
- Clinicians should recommend exercise and diet to older patients taking GLP-1 RAs

Clinical Relevance

GLP-1 receptor agonists (GLP-1 RAs) may accelerate muscle loss in older adults, but lifestyle modifications can help mitigate this effect. Clinicians should recommend exercise, dietary strategies, and routine body-composition assessments to optimize outcomes in older patients taking GLP-1 RAs.

Case Report

Patient 1

A 65-year-old woman came to our bariatric intake clinic with a weight of 127 kg and a BMI of 47.1. She had medically complicated obesity and multiple unsuccessful prior attempts at weight management. She expressed a desire to lose weight. Her medical history and current medications are shown in Table 1. She had attempted several lifestyle modifications over the past 25 to 35 years but had been unable to achieve weight loss. She engaged in limited physical activity, citing joint pain, exercise intolerance, and uncertainty about how to exercise safely as the primary barriers. Her typical food intake was relatively healthy and protein-focused; however, she frequently consumed restaurant meals, alcohol, and sweets (Table 2). Based on her typical daily meals and a daily protein recommendation of 1 g/kg body weight, her estimated average protein intake of 30 g/d was only 23.6% of her 127 g/d target (Table 1).[15]

Body composition was measured by using a segmental body composition analyzer (MC-580 scale; Tanita Corp) during the initial visit in July 2024 (Table 3). The patient was prescribed weekly 2-mL subcutaneous injections of tirzepatide, with doses that titrated from 2.5 mg/0.5 mL to 7.5 mg/0.5 mL over the observation period. Follow-up measurements, in November 2024, showed a 2.0% reduction in fat mass and a 15.1% reduction in lean mass. Because her fat mass remained relatively stable while her lean mass declined, her body fat percentage increased 3.5% over the observation period.

Patient 2

A 79-year-old woman came to our bariatric intake clinic with a weight of 76.8 kg and a BMI of 30.8. She had medically complicated obesity with multiple unsuccessful prior attempts at weight reduction and was seeking assistance with weight loss. Her medical history and current medications are shown in Table 1. She had previously tried dieting and commercial weight loss programs but experienced limited results. The patient reported taking 100 steps per day on a step-exerciser and occasionally hiking; she got at least 90 minutes of exercise per week. Her mobility was limited due to an L3-S1 spinal fusion, and she felt uncertain about how to safely exercise. Her typical food intake had a mix of carbohydrates, protein, and vegetables but also included occasional meals outside the home, alcohol, and sweets (Table 2). She consumed non–home-cooked meals 2 times per week. Based on her typical daily meals, her estimated average protein intake of 60 g/d was 78% of her recommended target of 76.8 g/d (Table 1).[15]

Body composition was measured during the initial patient visit in July 2024 with the same body composition analyzer used for patient 1 (Table 3). The patient was prescribed weekly semaglutide injections (pen injector), the dose of which was titrated from 0.25 mg/0.5 mL to 1 mg/0.5 mL over the observation period. Follow-up measurements, collected in January 2025, showed that the patient had lost 22.6% fat mass and gained 2.1% lean mass.

During an interview, patient 2 explained that she had used the mobile app MyFitnessPal to record her daily food intake since

Table 1
Clinical Characteristics of Each Patient

Characteristic	Patient 1	Patient 2
Age (years)	65	79
Diet quality	Low calorie, moderate protein (30 g/d)	Moderate calorie, moderate protein (60 g/d)
Race	White	White
Physical activity	Limited resistance training for upper back	Consistent at-home resistance exercises, 3×/wk for 15-20 min/session
Concurrent medications	Acyclovir, albuterol, duloxetine, fluticasone, indomethacin, lisinopril, lorazepam	Acetaminophen, albuterol, aspirin, beclomethasone, losartan, methocarbamol, metoprolol, rosuvastatin, levothyroxine
Comorbid conditions	Diffuse large B-cell lymphoma, right breast cancer, hypertension, osteoporosis, depression, obstructive sleep apnea	Hypertension, hyperlipidemia, hypothyroidism, osteopenia, thrombophilia (factor V Leiden, heterozygous), gastroesophageal reflux disease, prediabetes, chronic kidney disease (stage 3a)

Table 2
Typical Eating Habits of Each Patient

Category	Patient 1	Patient 2
Daily meals		
Breakfast	None or oatmeal	Honeycomb cereal with milk and coffee
Lunch	Sandwich or salad	Yogurt with fruit
Dinner	Turkey, fish, or chicken	Stuffed peppers with beef, along with rice and a salad
Weekly habits, frequency		
Eats food cooked outside home	4	2
Drinks liquor	4	4
Eats ice cream, cookies, or candy	5	5

Table 3
Body Composition Measurements Before and After Treatment With GLP-1 Receptor Agonists

Measurement	Patient 1		Patient 2	
	Before	After	Before	After
Weight, kg	127.0	116.1	76.8	70.8
BMI[a]	47.1	42.8	30.8	28.2
Body composition, kg (% of total)				
Fat mass	61.1 (48.1)	59.9 (51.6)	31.0 (40.4)	24.0 (33.9)
Lean mass	62.9 (49.5)	53.4 (46.0)	43.5 (56.6)	44.4 (62.7)

Abbreviations: BMI = body mass index; GLP-1 = glucagon-like peptide-1.
[a] Calculated as weight in kilograms divided by height in meters squared.

September 2024. She limited her caloric intake to 1200 kcal per day. Beginning in October 2024, she worked with a dietician to limit her sodium intake to less than 2300 mg per day and to include fruits, vegetables, and whole grains in her daily intake. Beginning in September 2024, she also followed a home gym routine for exercise, performing bodyweight exercises that target the abdominal, oblique, back, and arm muscles. She exercised 3 times per week for 15 to 20 minutes each session (Table 1). She attributed her substantial weight loss to snacking less frequently.

Discussion

This study shows that changes in body composition measurements can vary among older patients taking GLP-1 RAs. Any weight loss, whether from GLP-1 RAs or not, involves the loss of both fat mass and lean mass.[16] The loss of lean mass due to GLP-1 RAs can range from 15% to 60% of the total weight lost.[17,18] Lean mass loss varies due to factors such as age, disease state, amount of weight lost, and medication type.[19-22] On average, a person's relative lean mass percentage remains stable before and after GLP-1 RA treatment, suggesting that the amount of lean mass lost is proportional to the amount of total weight lost.[23] However, our case report shows that these trends do not apply to all patients. Lifestyle differences can lead to substantial variability in the amount of relative lean mass lost.

Our study showed 2 distinct patterns of weight loss in older women taking GLP-1 RAs. Patient 2's weight loss was almost entirely due to fat loss, with a gain in lean mass over the study period. In contrast, patient 1's weight loss was primarily from lean mass reduction. Notably, patient 1 was unable to maintain consistent physical activity due to significant pain in her hip and gluteal region as well as her back.

Given the importance of physical activity for preserving lean mass, the American Diabetes Association recommends that clinicians carefully evaluate a patient's risk of sarcopenia before prescribing GLP-1 RAs.[24] Maintaining consistent physical exercise while taking a GLP-1 RA may prevent sarcopenia in older patients. Home-gym and bodyweight routines may be more suitable for patients who cannot lift heavy weights. Additionally, the U.S. Department of Health and Human Services recommends that older adults partake in moderate-intensity aerobic and strength training exercise for at least 150 minutes per week; it also recommends that they prioritize a plant-based diet by eating fruits, vegetables, nuts, and grains.[25] The American Heart Association recommends the Mediterranean diet and the Dietary Approaches to Stop Hypertension diet to improve cardiovascular outcomes.[26]

Differences in medication type and duration of use may also contribute to variations in body composition changes. Our patients had different medication schedules, which may explain their differences in body composition during weight loss. Patient 1 used tirzepatide for 4 months, and patient 2 used semaglutide for 6 months. Tirzepatide is a dual RA for both GLP-1 and glucose-dependent insulinotropic polypeptide and generally results in faster weight loss than semaglutide does at similar time intervals.[27] Faster weight loss may also result in an increased rate of lean mass loss, which may partially explain why patient 1 had greater lean mass loss than patient 2. In comparison, the more gradual weight loss associated with semaglutide may enable the body to adapt metabolically to the medication and preserve lean mass.

Other factors, such as protein intake, other medications, and comorbid conditions, may also influence body composition changes. Patient 1 consumed a lower percentage of her recommended daily protein intake than patient 2, which may have contributed to her losing more lean mass. Patient 1 was taking duloxetine and lorazepam, both of which can cause fatigue and may have reduced the amount of physical activity she was able to undertake.[28] Patient 1 also had diffuse large B-cell lymphoma, which is associated with cancer cachexia and muscle wasting.[29] Furthermore, her depression and sleep apnea may have contributed to feelings of lethargy and decreased her likelihood of pursuing physical activity.[30]

Limitations of this study include the inclusion of only 2 cases, anecdotal evidence for differences in body compositions, and use of different GLP-1 RAs and study durations between patients. This case report is purely anecdotal and intended to suggest possible associations rather than establish causality.

Conclusion

Research that further investigates which lifestyle modifications can affect body composition changes will be crucial for tailoring therapies in older adults. We recommend that clinicians regularly measure body composition in older patients taking GLP-1 RAs to assess for signs of sarcopenia. When patients are losing more lean mass than fat mass, clinicians should evaluate whether a lack of physical exercise may be the cause.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used ChatGPT to format references. The authors also used OpenEvidence to find relevant references for scientific claims. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data Availability

All relevant data supporting the findings of this study are reported within the article.

Disclosure

The authors have no conflicts of interest to disclose.

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References

1. US Centers for Disease Control and Prevention. Adult obesity facts. Accessed March 25, 2025. https://www.cdc.gov/obesity/adult-obesity-facts/index.html
2. Cornier MA, Despres JP, Davis N, et al. Assessing adiposity: a scientific statement from the American Heart Association. *Circulation*. 2011;124(18):1996–2019. https://doi.org/10.1161/CIR.0b013e318233bc6a
3. Elmaleh-Sachs A, Schwartz JL, Bramante CT, Nicklas JM, Gudzune KA, Jay M. Obesity management in adults: a review. *JAMA*. 2023;330(20):2000–2015. https://doi.org/10.1001/jama.2023.19897
4. Powell-Wiley TM, Poirier P, Burke LE, et al. Obesity and cardiovascular disease: a scientific statement from the American heart Association. *Circulation*. 2021;143(21):e984–e1010. https://doi.org/10.1161/CIR.0000000000000973
5. Kalinkovich A, Livshits G. Sarcopenic obesity or obese sarcopenia: a cross talk between age-associated adipose tissue and skeletal muscle inflammation as a main mechanism of the pathogenesis. *Ageing Res Rev*. 2017;35:200–221. https://doi.org/10.1016/j.arr.2016.09.008
6. Wang T, He C. Pro-inflammatory cytokines: the link between obesity and osteoarthritis. *Cytokine Growth Factor Rev*. 2018;44:38–50. https://doi.org/10.1016/j.cytogfr.2018.10.002
7. Palma G, Sorice GP, Genchi VA, et al. Adipose tissue inflammation and pulmonary dysfunction in obesity. *Int J Mol Sci*. 2022;23(13). https://doi.org/10.3390/ijms23137349
8. Doyle ME, Egan JM. Mechanisms of action of glucagon-like peptide 1 in the pancreas. *Pharmacol Ther*. 2007;113(3):546–593. https://doi.org/10.1016/j.pharmthera.2006.11.007
9. Jastreboff AM, Aronne LJ, Ahmad NN, et al. Tirzepatide once weekly for the treatment of obesity. *N Engl J Med*. 2022;387(3):205–216. https://doi.org/10.1056/NEJMoa2206038
10. Wilding JPH, Batterham RL, Calanna S, et al. Once-Weekly semaglutide in adults with overweight or obesity. *N Engl J Med*. 2021;384(11):989–1002. https://doi.org/10.1056/NEJMoa2032183
11. Kitamura A, Seino S, Abe T, et al. Sarcopenia: prevalence, associated factors, and the risk of mortality and disability in Japanese older adults. *J Cachexia Sarcopenia Muscle*. 2021;12(1):30–38. https://doi.org/10.1002/jcsm.12651
12. Bluher M. Obesity: global epidemiology and pathogenesis. *Nat Rev Endocrinol*. 2019;15(5):288–298. https://doi.org/10.1038/s41574-019-0176-8
13. Kahan S, Manson JE. Obesity treatment, beyond the guidelines: practical suggestions for clinical practice. *JAMA*. 2019;321(14):1349–1350. https://doi.org/10.1001/jama.2019.2352
14. Apovian CM, Aronne LJ, Bessesen DH, et al. Pharmacological management of obesity: an endocrine society clinical practice guideline. *J Clin Endocrinol Metab*. 2015;100(2):342–362. https://doi.org/10.1210/jc.2014-3415
15. Cruz-Jentoft AJ, Volkert D. Malnutrition in older adults. *N Engl J Med*. 2025;392(22):2244–2255. https://doi.org/10.1056/NEJMra2412275
16. Weiss EP, Jordan RC, Frese EM, Albert SG, Villareal DT. Effects of weight loss on lean mass, strength, bone, and aerobic capacity. *Med Sci Sports Exerc*. 2017;49(1):206–217. https://doi.org/10.1249/MSS.0000000000001074
17. Neeland IJ, Linge J, Birkenfeld AL. Changes in lean body mass with glucagon-like peptide-1-based therapies and mitigation strategies. *Diabetes Obes Metab*. 2024;26(Suppl 4):16–27. https://doi.org/10.1111/dom.15728
18. Jiao R, Lin C, Cai X, et al. Characterizing body composition modifying effects of a glucagon-like peptide 1 receptor-based agonist: a meta-analysis. *Diabetes Obes Metab*. 2025;27(1):259–267. https://doi.org/10.1111/dom.16012
19. Wilkinson DJ, Piasecki M, Atherton PJ. The age-related loss of skeletal muscle mass and function: measurement and physiology of muscle fibre atrophy and muscle fibre loss in humans. *Ageing Res Rev*. 2018;47:123–132. https://doi.org/10.1016/j.arr.2018.07.005
20. Cohen S, Nathan JA, Goldberg AL. Muscle wasting in disease: molecular mechanisms and promising therapies. *Nat Rev Drug Discov*. 2015;14(1):58–74. https://doi.org/10.1038/nrd4467
21. Stefanakis K, Kokkorakis M, Mantzoros CS. The impact of weight loss on fat-free mass, muscle, bone and hematopoiesis health: implications for emerging pharmacotherapies aiming at fat reduction and lean mass preservation. *Metabolism*. 2024;161:156057. https://doi.org/10.1016/j.metabol.2024.156057
22. Dutt V, Gupta S, Dabur R, Injeti E, Mittal A. Skeletal muscle atrophy: potential therapeutic agents and their mechanisms of action. *Pharmacol Res*. 2015;99:86–100. https://doi.org/10.1016/j.phrs.2015.05.010
23. Karakasis P, Patoulias D, Fragakis N, Mantzoros CS. Effect of glucagon-like peptide-1 receptor agonists and co-agonists on body composition: systematic review and network meta-analysis. *Metabolism*. 2025;164:156113. https://doi.org/10.1016/j.metabol.2024.156113
24. Linge J, Birkenfeld AL, Neeland IJ. Muscle mass and glucagon-like Peptide-1 receptor agonists: adaptive or maladaptive response to weight loss? *Circulation*. 2024;150(16):1288–1298. https://doi.org/10.1161/CIRCULATIONAHA.124.067676
25. Piercy KL, Troiano RP, Ballard RM, et al. The physical activity guidelines for Americans. *JAMA*. 2018;320(19):2020–2028. https://doi.org/10.1001/jama.2018.14854
26. Fleg JL, Forman DE, Berra K, et al. Secondary prevention of atherosclerotic cardiovascular disease in older adults: a scientific statement from the American Heart Association. *Circulation*. 2013;128(22):2422–2446. https://doi.org/10.1161/01.cir.0000436752.99896.22
27. Rodriguez PJ, Goodwin Cartwright BM, Gratzl S, et al. Semaglutide vs tirzepatide for weight loss in adults with overweight or obesity. *JAMA Intern Med*. 2024;184(9):1056–1064. https://doi.org/10.1001/jamainternmed.2024.2525
28. Zhu M, Lv S, Zhu F, Zhang Y. Analysis of duloxetine-related adverse events using the food and drug administration adverse event reporting system: implications for monitoring and management. *J Clin Psychopharmacol*. 2025;45(2):96–105. https://doi.org/10.1097/JCP.0000000000001966
29. Baracos VE, Martin L, Korc M, Guttridge DC, Fearon KCH. Cancer-associated cachexia. *Nat Rev Dis Primers*. 2018;4:17105. https://doi.org/10.1038/nrdp.2017.105
30. Mjelle KES, Lehmann S, Saxvig IW, Gulati S, Bjorvatn B. Association of excessive sleepiness, pathological fatigue, depression, and anxiety with different severity levels of obstructive sleep apnea. *Front Psychol*. 2022;13:839408. https://doi.org/10.3389/fpsyg.2022.839408